IT World Essentials: Your Digital Entry Point

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Q.no1: Write problem statement.

Solution:
```
#include <stdio.h>

// User-defined function to calculate simple interest

float calculateInterest(float p, float r, float t) {

    return (p * r * t) / 100.0;

}

int main() {

    float principal, rate, time, interest;

    // Accept input from the user

    printf("Enter Principal: ");

    scanf("%f", &principal);

    printf("Enter Rate: ");

    scanf("%f", &rate);

    printf("Enter Time: ");

    scanf("%f", &time);

    // Calculate and display interest clearly

    interest = calculateInterest(principal, rate, time);

    printf("Simple Interest = %.2f\n", interest);

    return 0;
```

```
}
```

Q.no2: Write problem statement.

Solution:
```
#include <stdio.h>

// Function for total

float calculateTotal(float m1, float m2, float m3, float m4, float m5) {

    return m1 + m2 + m3 + m4 + m5;

}

// Function for percentage

float calculatePercentage(float total) {

    return (total / 500.0) * 100.0;

}

int main() {

    float m1, m2, m3, m4, m5, total, percentage;

     // Input marks from user

    printf("Marks: ");

    scanf("%f %f %f %f %f", &m1, &m2, &m3, &m4, &m5);

  total = calculateTotal(m1, m2, m3, m4, m5);

    percentage = calculatePercentage(total);

   // Display total and percentage

    printf("Total = %.0f", total);

    printf("Percentage = %.2f\n", percentage);
```

```
return 0;

}
```

Q.no3: Write problem statement.

Solution:

```
#include <stdio.h>

// User-defined function to check the PIN
void checkPIN(int pin) {
    // Compare it with predefined PIN (1234)
    if (pin == 1234) {
        printf("Access Granted\n");
    } else {
        printf("Invalid PIN\n");
    }
}

int main() {
    int enteredPIN;

    // Take PIN input from the user
    printf("Enter PIN: ");
```

```
    scanf("%d", &enteredPIN);


    checkPIN(enteredPIN);


    return 0;
}
```

Q.no4: Write problem statement.

Solution:

```
#include <stdio.h>


void checkPalindrome(int id) {
    int originalID = id;
    int reversedID = 0;
    int remainder;

    // Reverse the number using logic (loop)
    while (id != 0) {
        remainder = id % 10;
        reversedID = reversedID * 10 + remainder;
        id /= 10;
    }

    // Compare original number with reversed number
```

```
    if (originalID == reversedID) {
        printf("Palindrome\n");
    } else {
        printf("Not Palindrome\n");
    }
}

int main() {
    int employeeID;

    // Take employee ID as input from the user
    printf("Enter ID: ");
    scanf("%d", &employeeID);

    checkPalindrome(employeeID);

    return 0;
}
```

Q.no5: Write problem statement.

Solution: #include <stdio.h>

// Recursive function to generate Fibonacci numbers

```
int fibonacci(int n) {
    if (n == 0) {
        return 0;
    } else if (n == 1) {
        return 1;
    } else {
        return fibonacci(n - 1) + fibonacci(n - 2);
    }
}


int main() {
    int n, i;


    // Take input n from the user
    printf("n = ");
    scanf("%d", &n);

    // Print Fibonacci series up to n terms
    for (i = 0; i < n; i++) {
        printf("%d ", fibonacci(i));
    }
    printf("\n");


    return 0;
```

```
}
```